

APPENDIX K

CLAREMONT ROAD AND SIDEWALK CONSTRUCTION SPECIFICATIONS*

1. Road specifications.

A. *Grade stakes.* It shall be the responsibility of the developer to have grade stakes put in at fifty-foot intervals where there are abrupt changes in the terrain and on even one-hundred-foot stations where the land has a smooth contour. Each stake shall be driven firmly beyond the toe or top of the slope in a location where it will not be disturbed by construction operations and be clearly marked to give the following information:

1. Station
2. Offset from centerline
3. Cut or fill to finished centerline grade

The developer shall preserve the grade stakes until the completion of the streets and if stakes are removed or destroyed so that the representative of the city cannot readily check the grade at any location, the stakes shall be replaced at the expense of the developer.

Reference is made to the "Typical Roadway Sections" as minimum standards for roadway construction as supplemented by these specifications.

B. *Clearing and grubbing.* All trees and brush within the right-of-way lines or the slope lines, whichever is the farthest from the centerline, shall be cut, and either removed from the right-of-way

***Editor's note**—Printed herein are the City of Claremont Road and Sidewalk Construction Specifications adopted by the Claremont Planning Board on April 17, 1989. Style and capitalization have been made uniform. Obviously misspelled words have been corrected without notation. Words added for clarification have been added in brackets. Amendments have been included and are indicated by a history note immediately following the amended section.

or completely burned. Any partially burned wood shall be removed from the right-of-way.

All stumps and large roots shall be removed from all cuts and to a depth of four (4) feet below finished grade in fills. Stumps below five (5) feet shall be cut off within six (6) inches of the existing ground surface. Stumps and roots shall be disposed of outside the right-of-way.

Loam, forest litter, sod, muck, silt or other unsuitable material shall be removed below the limits of the roadbed to such depth as necessary to provide a stable sub-base.

Under no circumstances shall any wood, limbs, stumps, brush or other unsuitable material be placed under or allowed to remain within the limits of the roadbed.

C. Excavation and filled roadbeds. Excavation shall consist of removing earth or ledge to a depth of one and sixty one-hundredths (1.60) feet in clay and two and one-tenth (2.10) feet in marine soils below the finished grade shown on the final profile and to the approved typical section.

Filled roadbeds and embankments shall be formed by spreading successive layers not greater than six (6) inches in depth. Each layer shall be satisfactorily compacted to a density of at least ninety-five (95) percent of maximum density before another is begun.

Material containing loam, forest litter, large roots, stumps or other wood, muck, silt, clay or other substance which will not make a stable embankment will not be acceptable for the construction of fills.

Where filled roadbeds and embankments are to be made over ground that will not support construction equipment, the first layer may be made of a thickness of up to five (5) feet.

Ledge fragments or boulders larger than one-quarter cubic yard or eighteen (18) inches in diameter shall not be placed within two (2) feet of the subgrade.

D. *Gravel.*

1. *Bank run gravel.* Before spreading any gravel, the subgrade shall have been brought to the required elevation and cross section and thoroughly compacted, free from holes and ruts. Bank run gravel shall consist of the following gradations:

Twenty-five (25) to seventy (70) percent shall pass the No. 4 sieve and not more than twelve (12) percent of the material passing the No. 4 sieve shall pass the No. 200 sieve. No stones or rock fragments will be permitted which cannot be incorporated in a six-inch layer. Bank run gravel shall be placed in six-inch lifts and compacted to a minimum of ninety-five (95) percent compaction. A rubber tire roller and calcium chloride may be ordered upon the request of the appropriate city inspector in order to achieve compaction.

2. *Fine grading.* Fine grading will not be initiated until bank run gravel placing has been approved and not before forty-eight (48) hours to street paving unless all traffic can be kept off fine grading section until such time as street paving is done. All shoulder and ditch work, or any other excavation in the right-of-way shall be completed before final grading can begin.

a. *Required grading crushed gravel:*

<i>Sieve size</i>	<i>Percentage by weight passing</i>
3"	100
2"	95–100
1"	55–85
No. 4	27–52
No. 200 (Based on the fraction passing the No. 4)	0–12

Reference: Standard Specifications for Road and Bridge Construction, State of New Hampshire, Required Grading.

- b. *Fine grading procedure.* Prior to the start of fine grading, the contractor shall stake out the centerline of the road at one-hundred-foot intervals and at all centerlines of culverts and catch basins. Stakes shall be flagged in blue showing crushed gravel grade, and corresponding slope grade on side stakes. These stakes and grades shall be kept throughout fine grading operation in order to check proper alignment and crown in the road, and may not be pulled until ordered by the city inspector.

The contractor shall have the following pieces of equipment on the job before starting fine grading:

1. Grader with straight edge blade
2. Water truck
3. 8–12 ton steel roller
4. Pneumatic tire roller

Grades shall be checked and rechecked during final grading in order to get a uniform grade. The grades shall be maintained in front of the water truck and rollers in order to get grade and essential compaction. The city inspector will check crown and compaction at random points and will approve fine grading. Once approved, this grade shall be undisturbed until paved.

- c. *Street paving.* Street paving shall be accomplished by placing a minimum base course of two-tenths-foot of asphalt concrete three-quarters-inch to one-inch mix under State of New Hampshire specifications binder course. The wearing course shall be one-tenth (0.10) of asphalt concrete three-eighths-inch mix under State of New Hampshire Specifications Wearing Course. The two (2) courses will constitute a minimum compacted depth of three (3) inches. Materials, mixing, placing, compaction and related surfacing procedures are to conform to Plant Mix Pavements, General of Standard Specifications for Road and Bridge Construction State of New Hampshire. If in the opinion of the city inspector, there will be movement and settlement of the binder course, the wearing course will not be

applied until the following year. The time period between applying the binder course and the wearing course shall not exceed one (1) year.

- d. *Drainage.* Except in unusual circumstances, all future residential streets shall be constructed with underground storm drainage systems, as opposed to open ditching. Drop inlets shall be paved around grates, as shown in Typical Catch Basin Details.

Ditches and culverts where approved shall be installed as shown on the final plan, profile and/or typical sections as approved by the planning board. If during construction it develops that additional drainage pipes or structures are needed, the superintendent of public works and/or the city council maintain the right to have them installed before acceptance of the road.

All cross-culverts must be made of concrete or galvanized pipe.

No part of any culvert or drainage pipe shall be installed above subgrade level.

Driveway drains shall be installed if their omission will cause standing water in the normal ditch line.

Catch basins and other drainage structures shall be built to the standards of the New Hampshire Department of Public Works and Highways.

All pipes shall be laid upon a firm bed, true to line and grade, and back-filled tamped in layers not exceeding six (6) inches, in a workmanlike manner. Care shall be taken that the backfill material within six (6) inches of the pipe is a fine material which can be compacted. Each section of pipe shall be securely attached to the adjoining section by the approved method for the type of joint being used.

3. *Concrete curbs.*

- a. In general, the same construction specifications for sidewalks are to be followed for concrete curbs. The dimensions for concrete curbing being as follows:
 1. Eighteen (18) inches in height of which a six-inch finished face shall show and twelve (12) inches shall be below surface of street line.

2. Six (6) inches in thickness. Edges at top of curb shall be properly rounded with cement edger. Top of curb shall be flat and shall follow contour of street.
3. The base on which the curbing is to be placed shall be granular and well compacted.

Note: Granite or precast concrete curbing may be substituted for cast in place curbing provided that precast concrete curbing meets the same requirements as cast in place curbing.

b. Concrete specifications.

In all cases, the foundation shall be thoroughly moistened immediately prior to the placing of concrete. The proportioning, mixing and placing of the concrete shall be in accordance with Class A concrete as specified in the Standard Specifications for Road and Bridge Construction, State of New Hampshire, Department of Public Works and Highways. In addition, all concrete shall be air-entrained with the air content at seven (7) percent $\pm$ one (1) percent and have a slump of three (3) inches $\pm$ one (1) inch.

All concrete shall be ready-mixed and produced at a plant acceptable to the City of CLaremont. No hand or site mixing will be permitted.

Note: The concrete supplier shall certify that the concrete supplied meets the above noted specifications.

4. *Sidewalk construction.*

- a. Sidewalks shall have a minimum width of five (5) feet in residential areas. Sidewalks in business areas may vary from five (5) feet to fifteen (15) feet. Sidewalks are to be constructed so as to drain towards the gutter, the slope being continuous across the walk. The pitch shall be one-quarter inch to the foot. The walk shall be divided into five-foot sections to allow for expansion and contraction. Expansion joints with bituminous filler

shall be provided wherever new concrete adjoins other walks or rigid structures.

- b. All sod and vegetable matter shall be removed from construction site. Two-inch cushion of granular material such as sand must be provided and thoroughly tamped as a cushion for the sidewalk. All fills under sidewalks shall be of granular material, thoroughly compacted. The top of the fill shall be extended at least one (1) foot beyond the edges of the walks.
- c. Sidewalks abutting street surface shall be curbed. Sidewalks in park belts need not be curbed. Curbs shall be eighteen (18) inches in height of which six (6) inches of finished face shall be above the surface of the street.
- d. Sidewalk surfaces shall be slightly roughened to produce a nonslip surface. This shall be done by brushing the surface. All joints shall be edged with the proper tools.
- e. Concrete sidewalk shall be one-course construction and shall be covered after finishing, so as to insure proper curing. The cover shall be left on for a period of three (3) days and must be kept continuously wet. Membrane curing compounds may be used. They must be sprayed on immediately after finishing of concrete and must be sprayed on in an adequate and uniform coverage.
- f. Concrete shall not be poured on frozen ground or allowed to freeze. Calcium chloride shall not be used and no concrete shall be placed when the atmospheric temperature is below 35°F unless otherwise permitted.
- g. Concrete sidewalks shall have a uniform depth of six (6) inches except in driveways where the depth shall be a uniform eight (8) inches.
- h. Bituminous topping on sidewalks may be permitted in selected and approved areas, but these shall have either granite or concrete curbs.
- i. Sidewalk ramps, (curb cuts) are required wherever sidewalks on streets and ways or curbs are being constructed, reconstructed or other repair improvements

being made thereon, and they shall comply with the following:

Location. Sidewalk ramps shall be located at each corner of each intersection and also at each street crossing which is not an intersection. Sidewalk ramps at corners shall be as close as possible to the apex or the center of the radius of curb intersection with consideration for the intersection geometry and safety hazards.

When curbs or sidewalks are constructed or reconstructed on only one (1) side of the street, sidewalk ramps shall be installed on the opposite side(s) of the street. If driveways have side curbs, sidewalk ramps shall also be provided therein.

Slope of sidewalk shall not exceed one-in-twelve and slope shall blend to meet the roadway gutter with a one-half inch (13mm) maximum lip for drainage. Where sidewalks are too narrow to install a straight-line sidewalk ramp at a slope of 1-in-12, the sides of the sidewalk ramp shall also slope at one-in-twelve.

Width of sidewalk ramps shall be not less than forty (40) inches (1.02m) not including sloped sides. At any apex, no ramp shall exceed sixty (60) inches not including sloped sides.

The sides of sidewalk ramps shall extend no less than twenty-four (24) inches (.610m) at the curb. There shall be no vertical curbing at the side of the sidewalk ramp.

Curb height at areas other than ramps or transition shall not exceed seven (7) inches (.178m) nor be less than three (3) inches (.075m).

Texture of sidewalk ramp surface, including sloping, shall be roughened.

Indication that travel is not permitted shall be indicated by fences or railings forty (40) inches high (1m).

Approved by the Claremont Planning Board:

/s/ _____
 William Kimball, Chairman
 Date: April 17, 1989

MINIMUM GEOMETRIC AND STRUCTURAL GUIDES FOR LOCAL ROADS AND STREETS

Average Daily Traffic (veh./day)	0–200	200–400	400 and over
Pavement Width (feet)	20	22	24
Shoulder Width (feet)	4	4	8–10
Center of Road to Ditch Line	18	19–21	Varies
Pavement Type	Hot Bituminous	Hot Bituminous	Hot Bituminous
Slope of Roadway	¼" per ft.	¼" per ft.	¼" per ft.
Base Course			
Depth:			
(Gravel)	18"	20"	24"
(Cr. Gravel)	4"	6"	6"

- Notes
1. Grades of all streets not be less than one (1) percent and shall not exceed ten (10) percent.
 2. For Average Daily Traffic over 400 veh./day Paved Shoulders will be considered.
 3. Base Course Depths may need to be increased in areas of Poor Soils





